Three Ways the Published Values Can Reveal the Secrets

A step-by-step cryptanalysis of Chintamani, Paul, and Sa (2024)

Goal. We examine three different ways the secret-masking rule in the paper can reveal secrets. The order is: (1) one secret is leaked, (2) no secret is leaked but two public values are available, and (3) only one public value of the corrected numerical fixture is available.

What these attacks do not do. None of the attacks below breaks the elliptic-curve discrete logarithm problem (ECDLP). The ECDLP asks, in simple terms, whether one can recover a number a from a known curve point P and the point aP. The attacks here do not solve that problem or recover a hidden scalar from a curve point. Instead, they exploit the algebraic relationships between the values the scheme publishes, together with known constraints on the secrets and masks. The paper discusses the assumed difficulty of the ECDLP in its scheme description [1, Sec. 3.1, p. 1459]. Our attacks target the published masking formula, not the underlying ECDLP.

Scope of the results. The first two attacks follow from the paper's published-value formula under the conditions stated below. The third is demonstrated for the paper's small numerical example under the corrected extension-field encoding stated in Section 4. It is a demonstrated recovery for that fixture, not a general guarantee that one public value uniquely determines a secret; other valid inputs can leave multiple candidates. These are mathematical demonstrations of weaknesses in the specified construction; they do not, by themselves, establish whether these observations have been published previously.

1 The scheme's published values

The paper defines a bilinear pairing e and uses the property

$$e(P, iQ) = e(P, Q)^i.$$

Bilinearity is introduced in Section 2. In Section 3.1, the dealer defines

$$Q = \sum_j a_{j1} P, \qquad v_i = e(P, iQ) + K_i,$$

and publishes the values v_i [1, Secs. 2 and 3.1, p. 1459].

For convenience, name the common mask

$$z = e(P, Q).$$

Then the public-value formula becomes

$$\boxed{v_i = z^i + K_i.} \tag{1}$$

The secret K_i is an element of the prime field $\mathbb{F}_p$ (the paper chooses it from the integers $0, 1, \ldots, p-1$). The mask z is in the pairing's order-ℓ group μ_ℓ in an extension field. The paper discusses this field compatibility in Remark 2 of Section 3.1 [1].

The important point is that all the public values use powers of the same z.

2 One-key recovery

2.1 What does the attacker know?

Suppose the attacker learns the first secret K_1. This is the only secret the attacker needs to know at the start. The value v_1 is already public, so Equation (1) gives

$$v_1 = z + K_1.$$

Subtract the known secret from the public value:

$$\boxed{z = v_1 - K_1.} \tag{2}$$

Thus the common mask is recovered without reconstructing Q or computing a pairing.

2.2 Why does this reveal the other secrets?

For every other published value, Equation (1) says

$$v_i = z^i + K_i.$$

Rearranging gives

$$\boxed{K_i = v_i - z^i.} \tag{3}$$

The attacker knows v_i because it is public, and now knows z from Equation (2). They can therefore calculate every K_i.

Conclusion. If K_1 leaks, all the other secrets in this construction are exposed.

2.3 What if a different indexed secret leaks?

The same conclusion holds for a leaked K_t whenever t has a multiplicative inverse modulo ℓ. From the public v_t,

$$z^t = v_t - K_t.$$

Choose an integer d such that $td = 1 + a\ell$ for some integer a. Since $z \in \mu_\ell$, we have $z^\ell = 1$, and therefore

$$(v_t - K_t)^d = (z^t)^d = z^{td} = z^{1+a\ell} = z(z^\ell)^a = z.$$

So z can again be recovered. In the paper's example, $\ell = 37$ and there are four secrets, indexed 1 through 4. Each of those indices has an inverse modulo 37. For K_1, the simpler subtraction in Equation (2) is enough.

3 Two-value recovery

This attack needs no leaked secret. It uses the first two public values, together with the fact that each K_i lies in $\mathbb{F}_p$.

3.1 A field fact that removes the secrets

The field has characteristic p. In such a field,

$$(x + y)^p = x^p + y^p. \tag{4}$$

To see why, expand $(x + y)^p$ using the binomial theorem. Every middle term contains a binomial coefficient $\binom{p}{j}$ for $1 \le j \le p - 1$. Since p is prime, each such coefficient is divisible by p and is therefore zero in a field of characteristic p. Only x^p and y^p remain.

Also, $K^p = K$ for every $K \in \mathbb{F}_p$: this follows from Fermat's little theorem for nonzero K, and is immediate for $K = 0$.

3.2 Use the first public value

Write the first two public values as

$$v_1 = z + K_1, \qquad v_2 = z^2 + K_2.$$

Raise v_1 to the pth power and subtract v_1:

$$\begin{aligned} A &= v_1^p - v_1 \\ &= (z + K_1)^p - (z + K_1) \\ &= z^p + K_1^p - z - K_1 \\ &= z^p - z. \end{aligned}$$

The secret cancels. Everything on the left is public, so the attacker can calculate A.

3.3 Use the second public value

Do the same thing to v_2:

$$\begin{aligned} B &= v_2^p - v_2 \\ &= (z^2 + K_2)^p - (z^2 + K_2) \\ &= z^{2p} - z^2 \\ &= (z^p)^2 - z^2. \end{aligned}$$

From $A = z^p - z$, we get $z^p = z + A$. Substituting this into the expression for B gives

$$\begin{aligned} B &= (z + A)^2 - z^2 \\ &= z^2 + 2Az + A^2 - z^2 \\ &= 2Az + A^2. \end{aligned}$$

Rearrange:

$$\boxed{z = \frac{B - A^2}{2A}.} \tag{5}$$

This formula works when $A \neq 0$ and p is odd. Both conditions matter: if $A = 0$, this particular division is not possible.

3.4 When is A nonzero?

We found that $A = z^p - z$. Thus $A \neq 0$ exactly when $z^p \neq z$. The elements satisfying $x^p = x$ in the extension field are exactly the elements of $\mathbb{F}_p$.

Here is a short proof. Every element of $\mathbb{F}_p$ satisfies $x^p = x$, so the polynomial $X^p - X$ already has p roots. A nonzero polynomial of degree p cannot have more than p roots in a field. Consequently it has no other roots in the extension field. Therefore

$$\boxed{A \neq 0 \quad \Longleftrightarrow \quad z \notin \mathbb{F}_p.}$$

A useful sufficient condition comes from the pairing group. If $z \neq 1$ and $z \in \mu_\ell$ for a prime ℓ that does not divide $p - 1$, then $z \notin \mathbb{F}_p$. Otherwise z would lie in the multiplicative group $\mathbb{F}_p^*$, whose size is $p - 1$, and its order ℓ would have to divide $p - 1$. This is impossible under the stated condition.

The identity case can also be resolved when $\gcd(\ell, p - 1) = 1$. If $A = 0$, then $z \in \mathbb{F}_p^* \cap \mu_\ell$. This intersection has order $\gcd(\ell, p - 1)$, so in that case $z = 1$ and $K_i = v_i - 1$. The toy two-value script implements this branch. The BLS12-381 dealer excludes identity masks, and its attacker rejects $A = 0$ as outside that experiment's inputs.

Once z is recovered from Equation (5), calculate every secret using

$$\boxed{K_i = v_i - z^i.}$$

Conclusion. Under the conditions above, two public values are enough to recover the mask and then all the secrets. No secret needs to leak first.

4 One-value recovery: the corrected numerical fixture

This demonstration uses only the first public value v_1 and the known structure of the mask. We work through the paper's numerical example under the corrected encoding, where $p = 47$, the mask lies in μ_{37}, and $K_1 = 8$ [1, Sec. 5, pp. 1462–1463].

4.1 What is known?

The displayed public value is

$$v_1 = 37\alpha^2 + 28\alpha + 12,$$

where the working Sage reproduction represents the extension field using

$$\alpha^3 + 3\alpha + 42 = 0.$$

A note on the numerical representation appears at the end of this section.

The attacker knows v_1, but not K_1. They try a candidate $k \in \mathbb{F}_{47}$ for the secret. This candidate would imply the mask

$$w = v_1 - k.$$

For this to be the real mask, it must satisfy

$$w^{37} = 1, \tag{6}$$

because the mask belongs to the order-37 group μ_{37}.

Checking Equation (6) for all 47 possible values of k would already be a finite, direct test for this small example. We can first narrow the candidates using a cubic equation.

4.2 A necessary condition: the norm

For an element w in this degree-three extension, define its norm by

$$N(w) = w\, w^{47}\, w^{47^2}.$$

The three factors are the element and its two Frobenius conjugates. Their product lies in $\mathbb{F}_{47}$; raising the product to the 47th power simply cycles the three factors.

If w is a valid mask, then $w^{37} = 1$. Hence

$$\begin{aligned} N(w) &= w^{1+47+47^2} \\ &= w^{2257} \\ &= (w^{37})^{61} = 1, \end{aligned}$$

where $2257 = 1 + 47 + 47^2 = 37 \cdot 61$. So every valid mask must satisfy $N(w) = 1$.

For the candidate $w = v_1 - k$, the condition is

$$N(v_1 - k) = (v_1 - k)(v_1^{47} - k)(v_1^{47^2} - k) = 1. \tag{7}$$

The substitutions $k^{47} = k$ and $k^{47^2} = k$ are valid because $k \in \mathbb{F}_{47}$. In the stated field representation, direct field arithmetic gives

$$\begin{aligned} v_1^{47} &= 22 + 27\alpha + 42\alpha^2, \\ v_1^{47^2} &= 15 + 39\alpha + 15\alpha^2. \end{aligned}$$

Multiplying the three factors in Equation (7), and reducing all coefficients modulo 47 using $\alpha^3 = -3\alpha - 42$, gives

$$N(v_1 - k) = 12 - k^3 + 2k^2 - 18k.$$

Set this equal to 1 and rearrange:

$$k^3 - 2k^2 + 18k - 11 = 0 \pmod{47}.$$

The cubic factors as

$$\boxed{(k - 8)(k - 12)(k - 29) = 0 \pmod{47}.}$$

Thus the norm condition leaves three candidates:

$$k \in \{8, 12, 29\}.$$

This condition is necessary, but it is not sufficient: it tells us the candidate mask has norm 1, not yet that its 37th power equals 1.

4.3 Check which candidate is a valid mask

Now test the original requirement $w^{37} = 1$ for each remaining candidate. Using the same field representation gives

k	$(v_1 - k)^{37}$
8	1
12	$42 + 5\alpha + 41\alpha^2$
29	$18 + 17\alpha + 9\alpha^2$

The last two results are not 1: in this field representation, an element equals 1 only when its constant coefficient is 1 and its α and α^2 coefficients are both zero. Therefore $k = 12$ and $k = 29$ are rejected, leaving

$$\boxed{K_1 = 8.}$$

The attacker has recovered the first secret from one public value and the mask's mathematical constraints. This unique result is established for this numerical example. Other valid public values can leave multiple keys even after the subgroup test, including within the same field and subgroup. The implementation reports all surviving candidates rather than assuming uniqueness.

Representation note. Section 5 of the paper gives $X^3 + X + 4$ as an example of an irreducible cubic, while the working Sage reproduction used for the displayed point and value calculations uses $X^3 + 3X + 42$. The coefficient calculations above are for the latter representation. Coordinates and coefficients tied to one defining polynomial should not be mixed with another without converting the representation [1, Sec. 5, pp. 1462–1463].

5 Summary

Attack	Attacker starts with	Main step
One-key recovery	One known secret K_1	$z = v_1 - K_1$
Two-value recovery	Public v_1, v_2	Remove secrets with pth powers
One-value fixture	Public v_1 in the corrected fixture	Norm gives 3 candidates; test $w^{37} = 1$

All three analyses use the published relationship $v_i = z^i + K_i$. The first is a consequence of one secret being compromised. The second shows how two public values can be enough under a stated field condition. The third uses additional structure in the paper's particular small numerical example.

6 Current repository implementation

The main script, `chmss_two_level_demo.sage`, constructs and reconstructs point shares through Levels 1 and 2 and recovers one secret, $K_1 = 8$. Its fixed toy fixture uses $Q = 7P$. It does not implement the full three-level example or submitted-share verification.

The attack directories implement the masking layer separately:

- `attack/one_key/toy_model/`: two toy keys with $Q = 7P$; the attacker uses the fixed simulated leak `LEAKED_K1=8`.
- `attack/two_values/toy_model/`: four toy keys with $Q = 7P$; the attacker receives no compromised key.
- `attack/one_value/`: only the first published value of the corrected $Q = 13P$ fixture. The attacker enumerates all 47 possible keys, tests their candidate masks' norms, then applies $w^{37} = 1$. It reproduces the candidates $8, 12, 29$ and the unique survivor 8 calculated in Section 4, without reading the expected key. This demonstrates recovery for the corrected numerical fixture; other valid inputs can leave multiple candidates.
- `attack/one_key/bls12_381/` and `attack/two_values/bls12_381/`: four random 256-bit keys embedded in the BLS12-381 prime field, using standard G1/G2 generators and a reduced Tate pairing. These examples instantiate the same masking relation, rather than the complete hierarchical protocol.

Each experiment has `dealer.sage`, `attacker.sage`, and a `public_values.txt` transcript beside the scripts. Toy one-key and two-value transcripts use comma-separated power-basis coefficients; the one-value transcript contains three space-separated coefficients. The BLS transcripts use JSON. The public files contain masked values, not private keys or masks. Dealers overwrite their own public transcript. The BLS one-key dealer also writes a separate `dealer_truth.json` record for verification and prints the first key to model a compromise.

All experiment scripts run directly from the repository root as `sage <path>.sage`, with no command-line flags or prior Sage session. The BLS one-key attacker prompts for the compromised first key. Its verifier supplies that input automatically. Both `curve.py` helpers can also be executed directly with Sage; they define functions rather than run an attack. Paths and helper imports resolve relative to the script or module. Commands and full input details are in `attack/README.md`.

Only the BLS directories include `verify_attack.sage`. Each compares all four recovered keys with dealer ground truth in two fresh trials. The two-value verifier uses separate namespaces within one process; the one-key verifier uses Sage subprocesses and moves the private ground-truth record away before supplying only K_1 to the attacker. It deletes that record during cleanup. Neither mechanism is an operating-system sandbox. The toy directories have no separate ground-truth verifier; the one-value candidate set establishes consistency with the public constraints, and a singleton set establishes unique recovery under those constraints.

References

[1] M. Chintamani, P. Paul, and L. Sa, "Conjunctive Hierarchical Multi-Secret Sharing Scheme using Elliptic Curves," *Indian Journal of Pure and Applied Mathematics*, vol. 55, pp. 1456–1464, 2024. doi:10.1007/s13226-023-00450-x.